

A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs

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PC-Max | Better preference scores through time-aware, parallel search

01 MOTIVATION

Fastest routes need not be the **safest**, most **navigable**, or most **scenic**.

Maximize a user preference while keeping travel time within an acceptable upper bound.

This computationally hard problem reduces to an instance of the NP-hard Arc Orienteering Problem [1].

The challenge: a large search space, time-dependent costs, compatible subpath timings, and a real-time response deadline.

02 PROBLEM DEFINITION

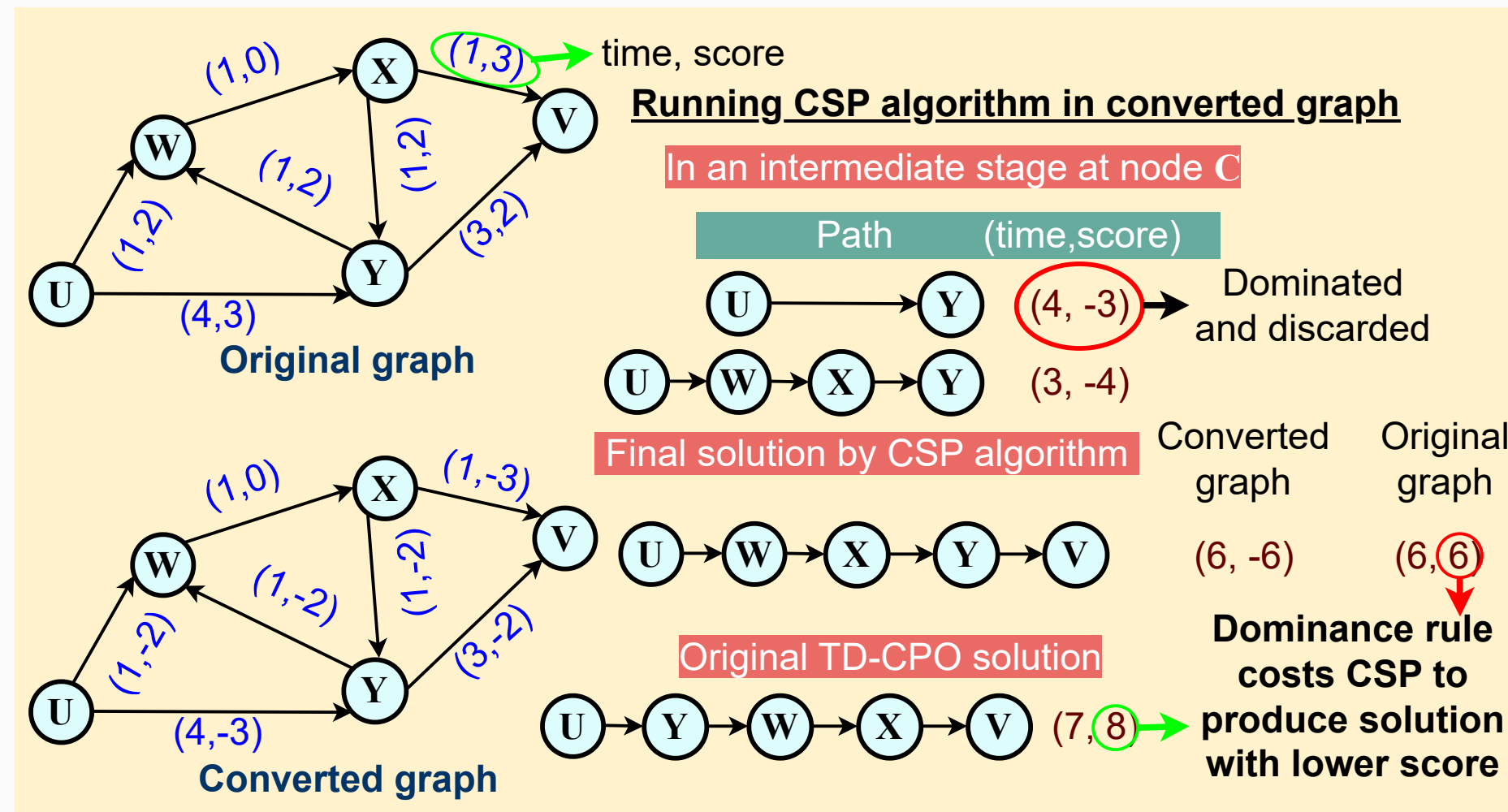
Input: time-dependent graph $G = (V, E)$; source s , destination d ; departure t_{dep}^s ; travel-time budget B ; edge preference $\Phi(e, t)$.

$$\begin{aligned} &\text{maximize} && \Phi(\mathcal{P}, t_{dep}^s) \\ &\text{subject to} && \Gamma(\mathcal{P}, t_{dep}^s) \leq B. \end{aligned}$$

Φ : **preference score**. Γ : **elapsed travel time**.

Return a **loopless** s – d path; no waiting at intermediate nodes. Edge scores are evaluated at their departure times. FIFO travel is assumed.

03 WHY STANDARD CSP FALLS SHORT



Negating scores and applying standard CSP dominance can discard useful partial paths: different visited vertices permit different loopless continuations.

Sequential search also limits parallelism. PC-Max instead exposes independent edge–pivot subproblems.

04 DATASETS & EXPERIMENTAL SETUP

~24M / ~58M nodes / edges

Road network	Nodes	Edges
Oldenburg (OL)	6,105	7,035
New York City (NY)	264,346	733,846
Florida (FLA)	1,070,376	2,712,798
California + Nevada (CAL)	1,890,815	4,657,742
USA	23,947,347	58,333,344

Real road networks; synthetic attributes. Travel-time profiles model morning/evening rush hours; piecewise preference functions are assigned to randomly selected edges.

3-SECOND RUNTIME LIMIT for PC-Max

Parameter	Default	Tested
Overhead	30%	20, 30, 40, 50%
Non-zero-score density	20%	10, 20, 30%
Threads	1	1, 6, 12, 24, 48
Recursion depth	2	1, 2, 3
Departure	Rush hour	Also outside

Overhead is measured relative to fastest-path travel time; B is that travel time plus overhead.

Java · Intel Xeon Gold 6258R

28 cores / 56 threads; 360 GB RAM; 2.7 GHz base / 4.0 GHz maximum boost.

Large-network plots compare A*, REC-INSERT [2] and SCOPE [4] with PC-Max on a **single core**. SCOPE also has a 3-second cap. PCSP and MEMETIC [3] appear only in the small OL comparison; the paper also evaluates Adv_Fn.

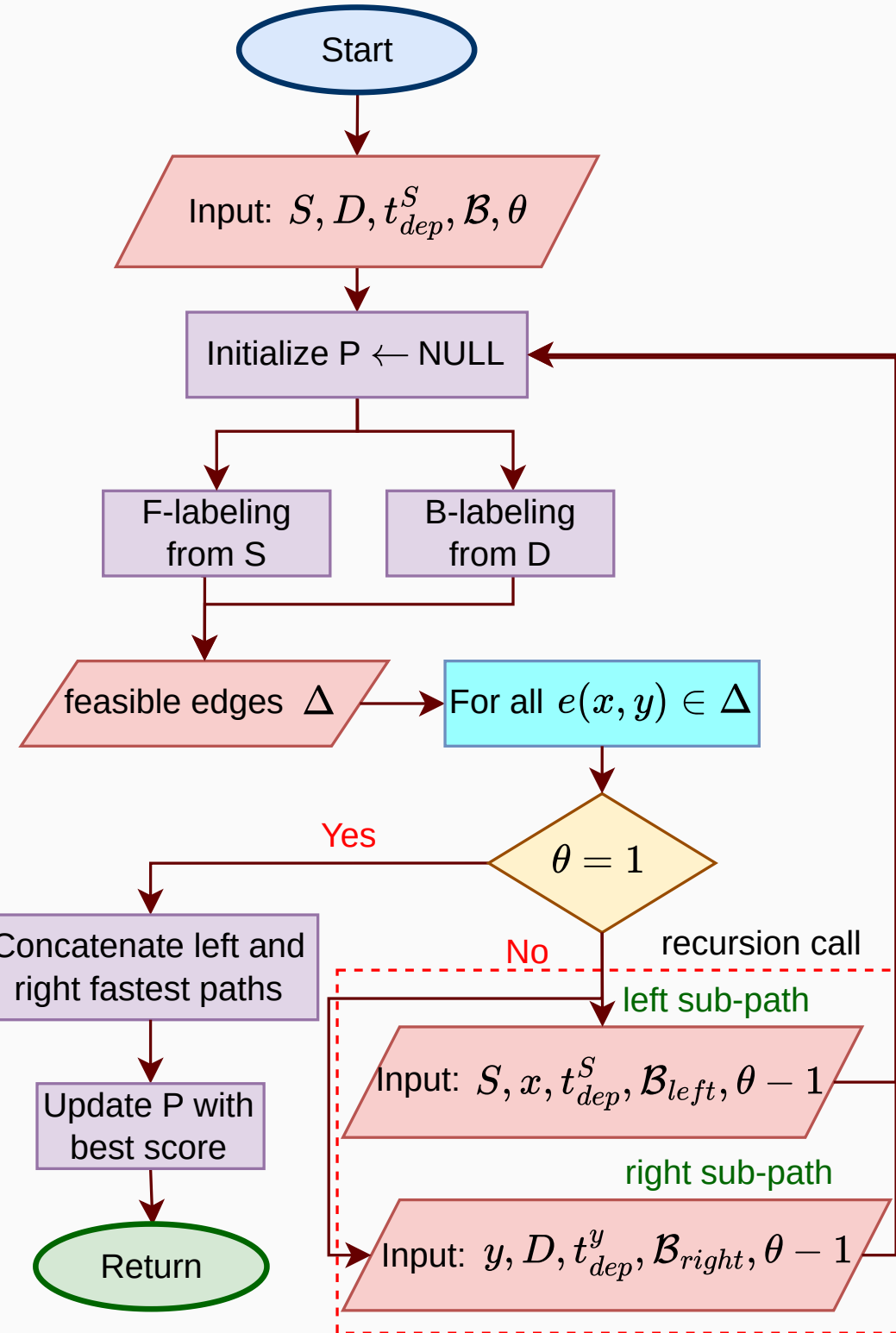
Search-space behavior. Larger budgets expand $|\Delta|$, pivot ranges and recursive calls. More overhead or non-zero-score edges increases achieved scores.

SELECTED REFERENCES

- [1] D. Gavalas et al. *Approximation algorithms for the arc orienteering problem*. Information Processing Letters 115(2), 313–315 (2015).
- [2] Y. Lu et al. *Scenic routes now: Efficiently solving the time-dependent arc orienteering problem*. CIKM, 487–496 (2017).
- [3] C. Chen et al. *Enjoy the most beautiful scene now: a memetic algorithm to solve two-fold time-dependent arc orienteering problem*. Frontiers of Computer Science 14(2), 364–377 (2019).
- [4] K. K. Dutta and V. M. V. Gunturi. *Constrain Path Optimization on Time-Dependent Road Networks*. WISE 2024 PhD Symposium, Demos and Workshops (2024).

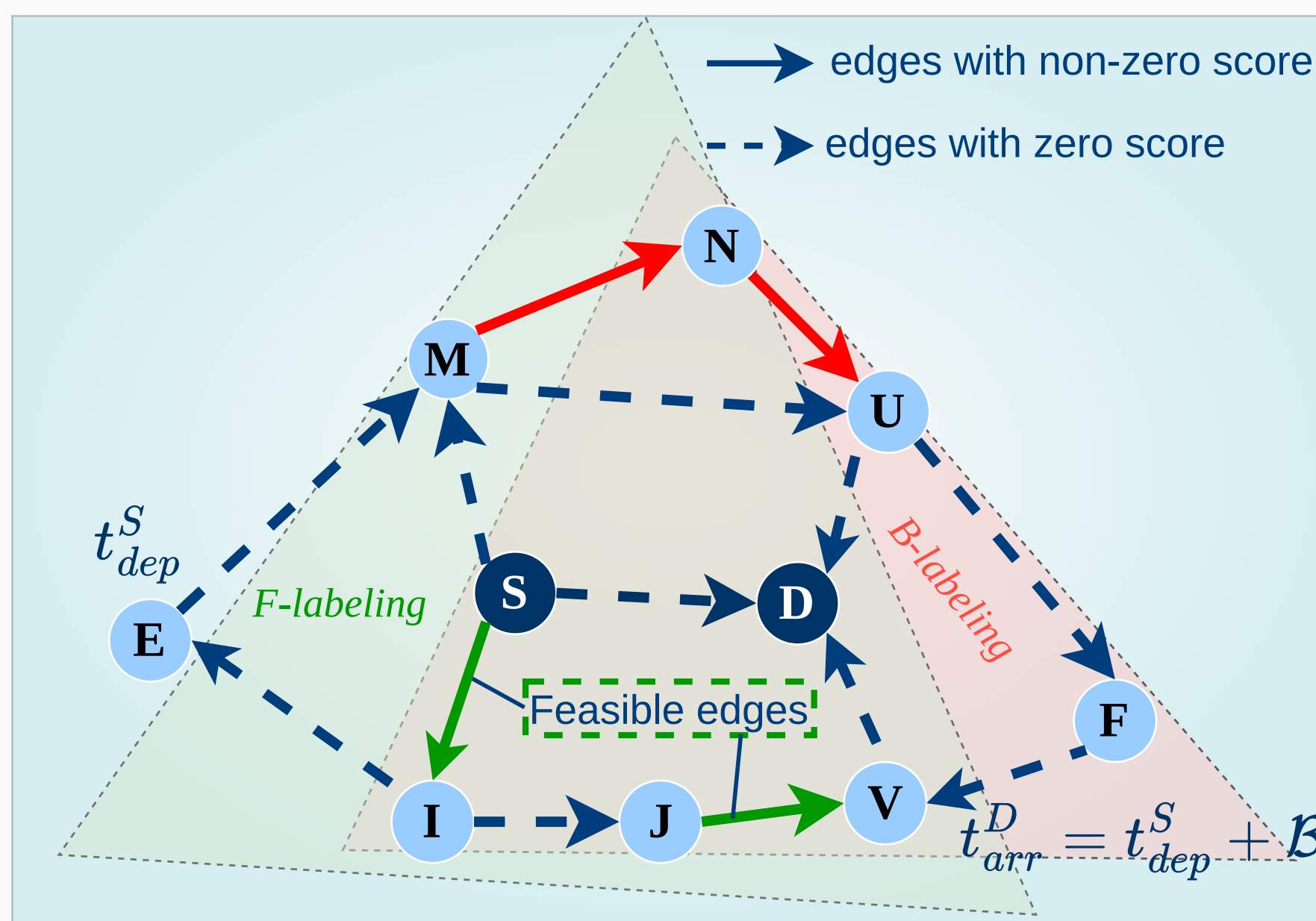
05 PC-MAX OVERVIEW

Filter → decompose → align → select.



F- and B-labeling identify Δ . Recurse on left/right subpaths for each edge and pivot; join compatible paths and keep the best feasible, loopless candidate.

06 FEASIBLE EDGE SELECTION



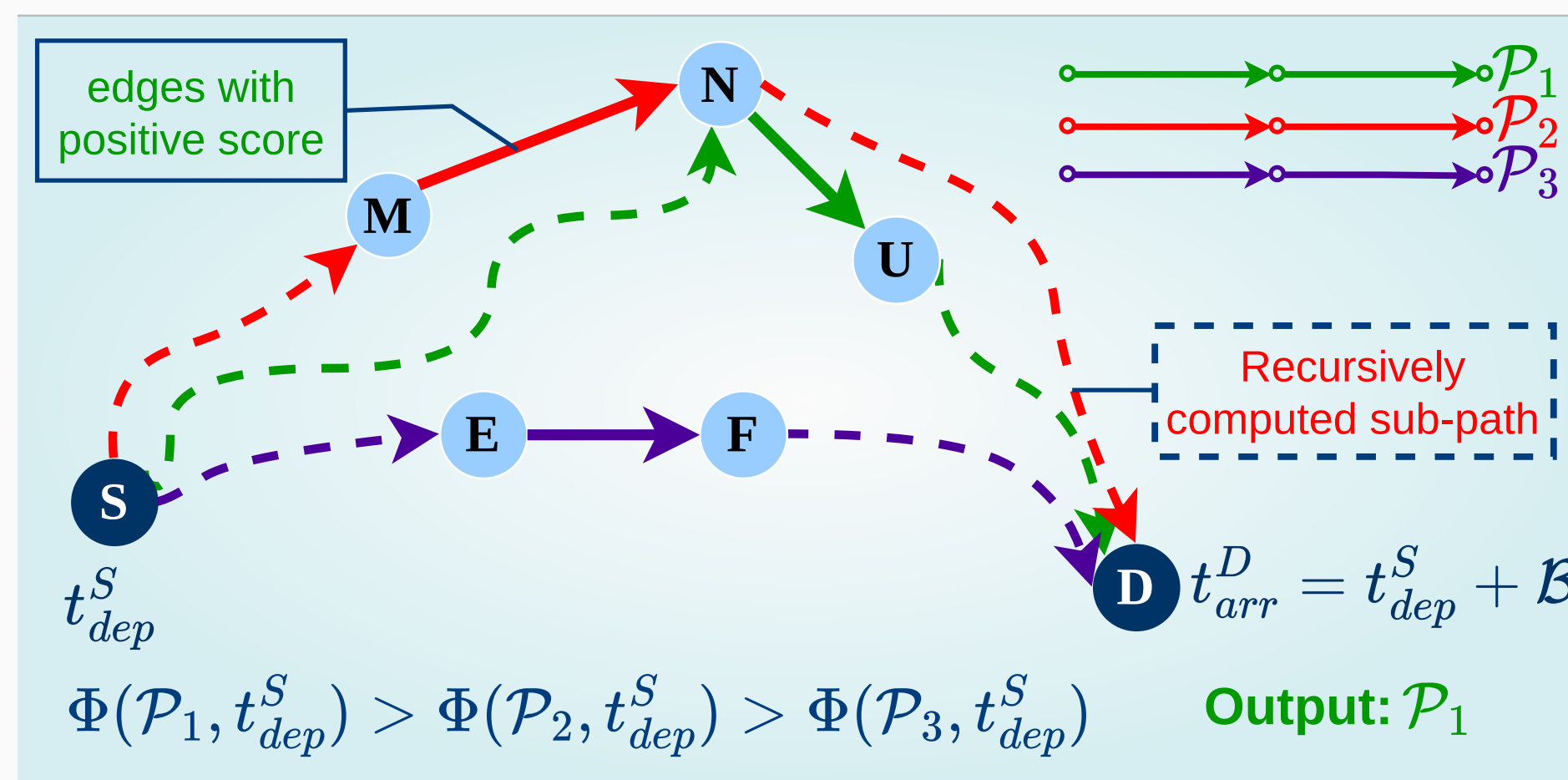
F-labeling: earliest arrivals from the source.

B-labeling: latest departures that still reach the destination by $t_{arr}^d = t_{dep}^s + B$.

The overlap, timing checks and non-zero scores define Δ , retaining all objective-contributing edges of a feasible loopless path.

Search within the budget-feasible region.

07 EDGE-PIVOT RECURSION



$$\underbrace{u \rightsquigarrow x}_{\mathcal{P}_{left}} \xrightarrow{e(x,y)} \underbrace{y \rightsquigarrow v}_{\mathcal{P}_{right}}$$

Pivots range from earliest arrival at x to the latest departure across e that permits timely arrival at v . Each fixes the left budget, right departure and remaining budget.

TEMPORAL ALIGNMENT

Match the left path's **actual arrival** at x (LP-MAP key) with the right path's **pivot** (RP-MAP key). Then stitch without waiting; validate budget and looplessness.

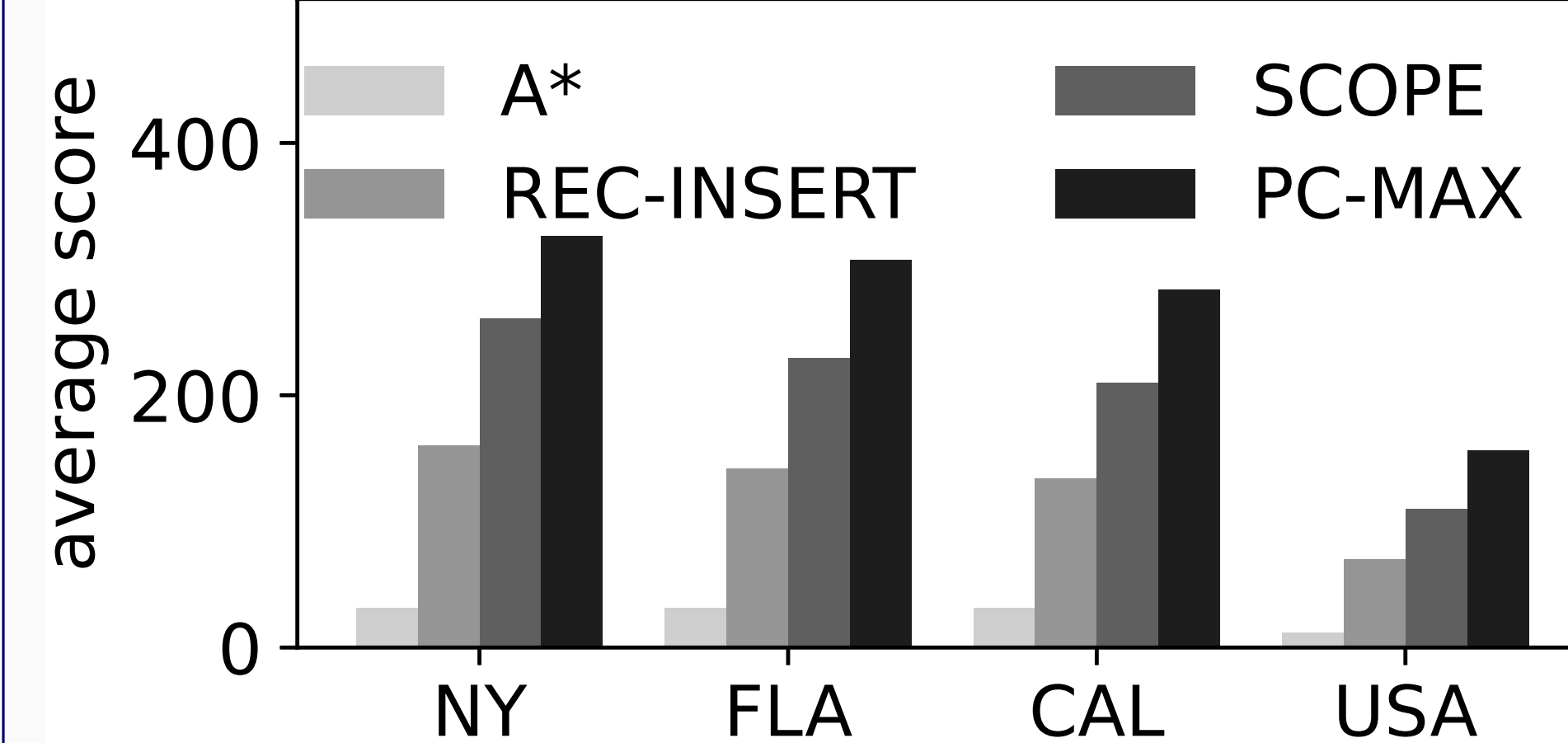
08 REUSE & PARALLEL EXECUTION

Reuse recursive results. Share maximum-budget F-/B-labeling across recursive calls; store subpaths in LP-MAP / RP-MAP.

Feasible edges × discrete pivots
↓ independent subproblems ↓
Work-stealing / fork-join execution
↓ align, validate, select best

09 HIGHER PREFERENCE SCORES

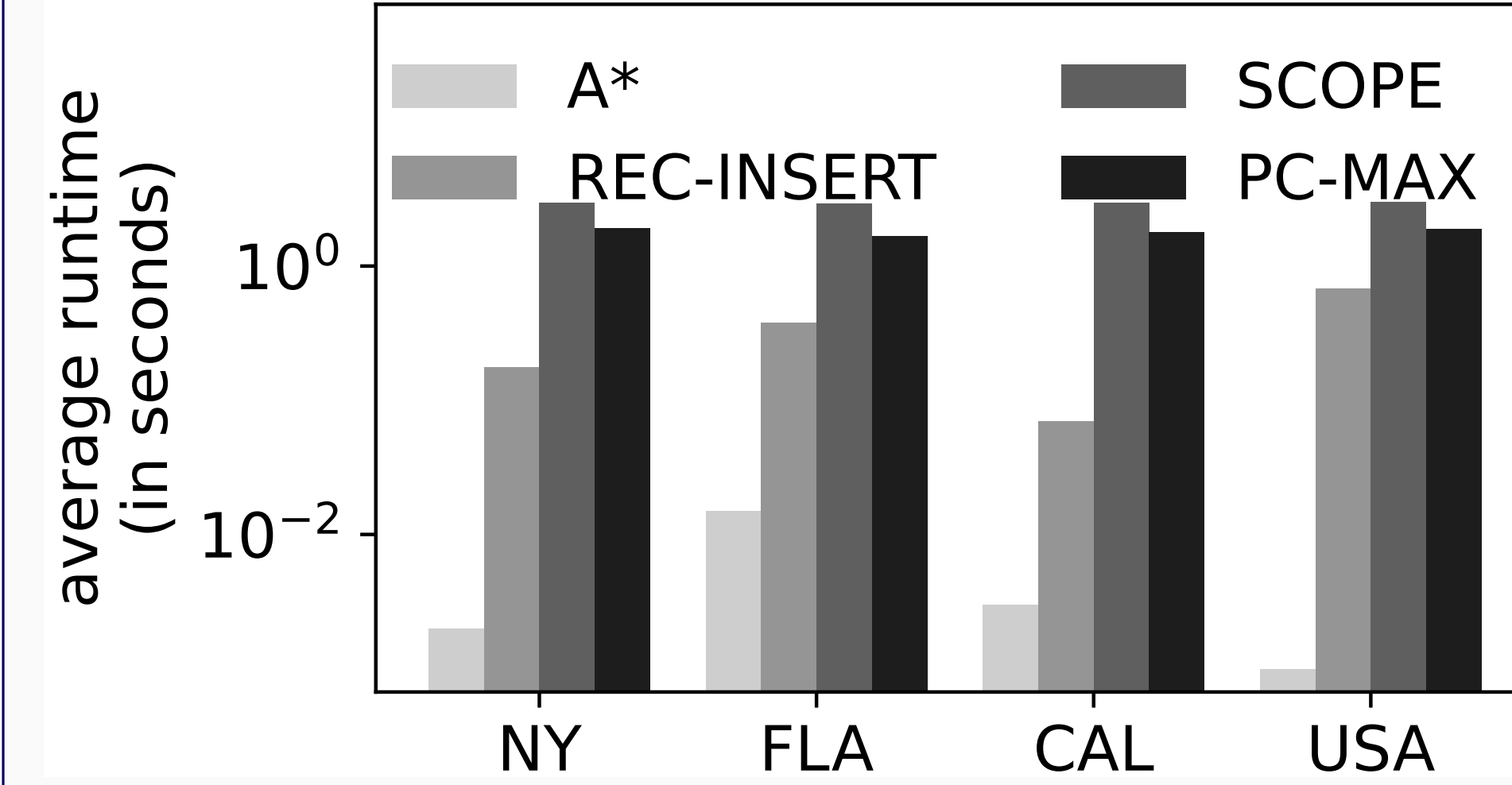
Highest reported scores on the large networks.



Score comparison · single core. A*, REC-INSERT, SCOPE and PC-Max.

≥ 1.29 × SCOPE score in USA

Up to 1.45 × SCOPE score in FLA; average 1.7 × REC-INSERT score in USA; up to 15 × A* score.



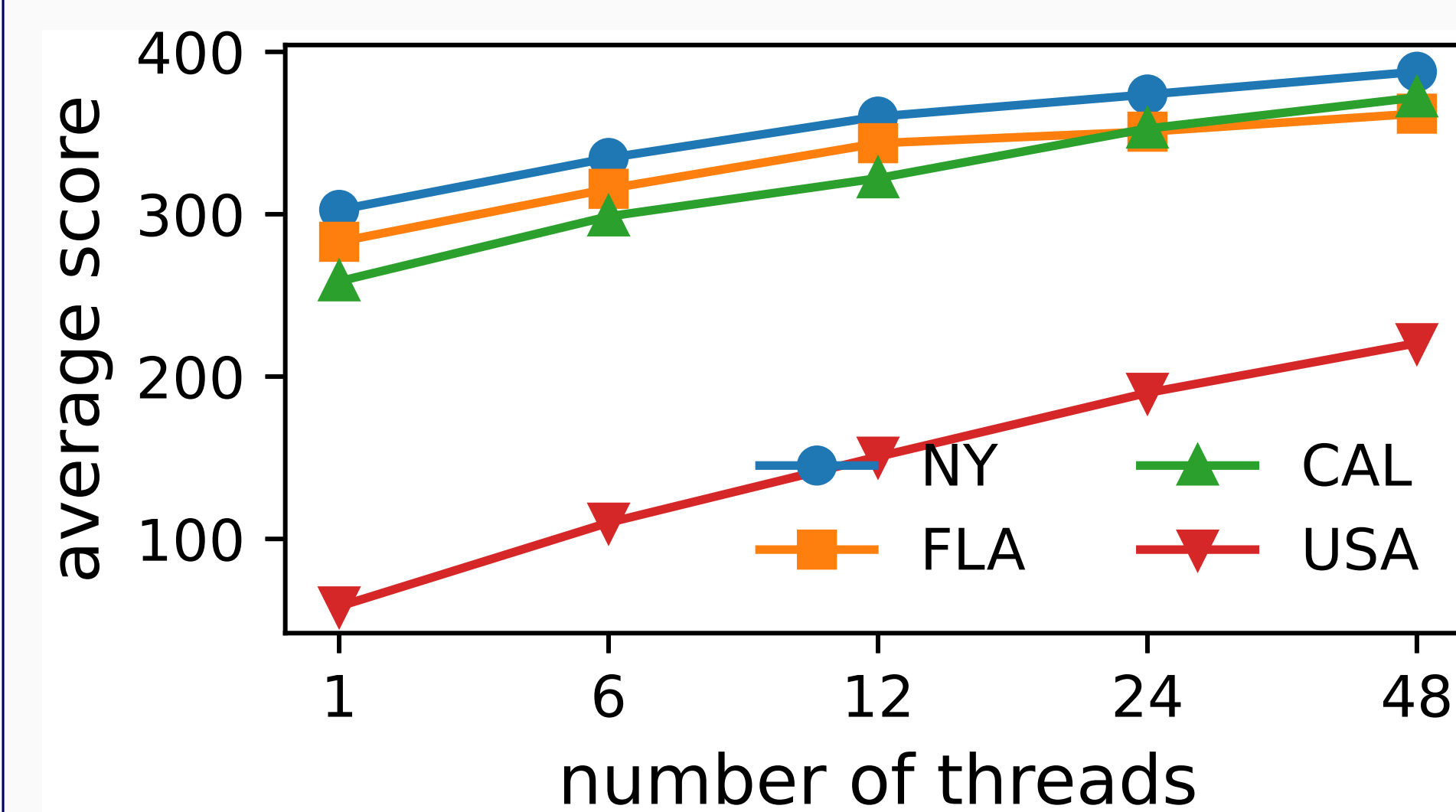
Runtime · single core · log scale.

1.92 s up to, on USA in this comparison

10 MORE THREADS, BETTER ROUTES

3.8 × up to, preference-score improvement

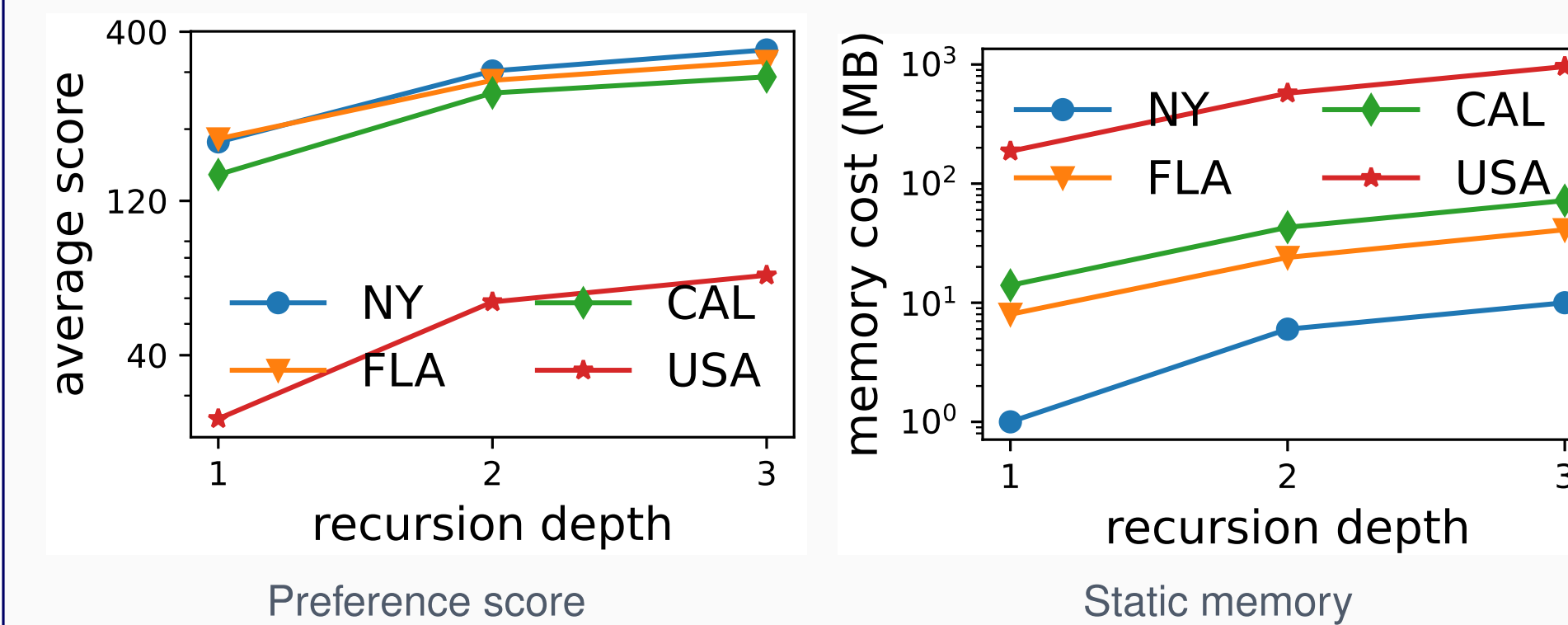
48 threads versus 1 | same 3-second limit



More threads evaluate more independent edge–pivot combinations before the deadline, improving the score.

Queries here have fastest-path durations of 51–60 minutes. The plotted metric is score under a fixed runtime budget.

11 PRACTICAL RECURSION DEPTH



$\theta = 2$ captures most observed quality gain. Deeper search raises cost sharply; bounded depth does not guarantee optimality.

At $\theta = 3$, USA static memory peaks at ~960 MB, excluding the graph and runtime thread allocations.

The paper's exhaustive-depth result uses $\theta = |N|$, where N contains edges with positive score at some time in the query horizon.

12 TAKEAWAYS

Higher-quality routes through a richer search of the feasible region.

Time-aware decomposition aligns recursive subpaths before combining them.

Parallel by design improves scores within a fixed execution deadline.

Practical at scale bounded depth, reused results, networks up to ~24M nodes.

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